IMPLEMENTING FHIR® WITH COMPOSABLE AUTOMATION™

FHIR and HL7® bring dramatic improvements in the interoperability of healthcare systems; using a Composable Automation™ Platform to implement HL7 accelerates that implementation and provides great flexibility for its ongoing growth and expansion.

HL7 IS THE FOUNDATION

The HL7® standards for sharing healthcare information have been in development since 1987, enabling faster and more reliable communication of vital information among providers, payers, government agencies, electronic medical record systems, etc. More timely, complete, and accurate information improves patient care and provider effectiveness, which also helps reduce the total cost of healthcare for all participants.

The name HL7 refers to "Health Level Seven® (HL7) International", a non-profit organization with members around the globe working to improve healthcare interoperability.

FHIR ADDS SPEED AND EASE OF IMPLEMENTATION

FHIR® (pronounced "fire") adapts the HL7 work to the newest methods of software development and connectivity. As the HL7 organization explains, FHIR® "combines the best features of existing HL7 standards with the latest web technologies to make interoperable healthcare applications dramatically simpler, easier, and faster to develop."
As a result, you can build on the FHIR® standards using software development skills, tools, and data formats your team already understands.

However, you should think about FHIR®, the "Fast Healthcare Interoperability Resources" standard as a platform, not a solution. It is a specification on which many solutions are being built. It defines a framework, and the ability to connect effectively with many other entities to accomplish the specific data exchange goals your business needs to reach.

PUTTING IT INTO ACTION

What is required to build interconnections using HL7 and FHIR®? There are hundreds of "implementation guides" defining the requirements for well-understood data representations and exchange in different situations and countries. As noted, an enormous variety of solutions can be built using the framework and standards, and each particular application you build will have somewhat different implementation steps.

Given the complexity hidden behind the simple phrase "healthcare information," the HL7 specification, though large, is intended to be as simple and clean as possible, focused only on the most common needs. The HL7 organization expects the more unusual requirements to be handled by extensions of the core standard. Your particular needs might lead you to adopt or create such extensions, which are offered to the general community as "packages".

The exchange of information between providers, payors, health information exchanges (HIE), government agencies, care coordination platforms, etc. may take several forms. Quick, interactive requests and responses can be handled by the normal internet tools commonly used with every access to a web page. Other uses will be better served by classic messaging approaches or document transfers.

HEALTHLX AND TESCHGLOBAL

The HealthLX team brings expertise in building automated workflows on the HL7 and FHIR® foundation.

For example, we provide a fully managed solution to satisfy all new mandates for the CMS Patient Access final rule (CMS-911-F), and stand ready to help organizations satisfy the newly proposed rule CMS-0057-P for interoperability and prior authorization.

We provide seamless integration with Zyter|TruCare standards in support of HL7 2.x ADT transactions to enable more prompt communication of patient care activity, We enable the translation of legacy healthcare data assets such as Continuity of Care Documents (CCDA), to FHIR® resources.

Our Composable Automation ™platform provides for API orchestration with best of breed applications and legacy systems in support of complex workflows, and enables the import of complex Data Warehouse information for use in enterprise analytics.

Our companion organization, TeschGlobal, has many years of experience solving the complex, continually evolving problems of integrating computer systems among trading partners across multiple industries. The varieties of connection standards, data formats, and 'systems of record' in the world sometimes seem endless. The team's ability to knit together information flows across those barriers is a valuable skill only gained from long practice.

Adding to that, healthcare brings its own collection of EMR systems, compliance challenges, and syntax issues, among others, and compliance deadlines add urgency to the task. With a rich history of solving complex data flow challenges combined with healthcare domain knowledge, the two organizations can provide “custom fit” enterprise solutions based on unique client needs.

IMPLEMENTATION

Our approach to planning, delivery, and ongoing support ensures the success of new integrations.

We address solution design, configuration and development, training and education, data transfer, sizing, system migration, change management, customer support, and ongoing managed services as desired, all based on proven best practices reinforced by experience spanning nearly 20 years.

In the process, we cover issues such as the data backbone, EMR integration, user interfaces, visualization and reporting, interactive forms, etc.
Any implementation will require the normal project phases to be done well, including all of the following activities:

- ✓ The design of the solution will include careful mapping of the existing data to the HL7 standard, in both directions, so data can be sent to, and received from any HL7-compliant repository or trading partner, and understood correctly on both sides.
- ✓ The connections will be integrated with your existing business processes, to support and extend the value you've already built.
- ✓ The components of the process will be designed to be reusable in other flows, and other applications not yet being built, in line with our philosophy of "Composable Thinking".
- ✓ Preparation for future scalability and high availability needs will be carefully planned.
- ✓ Testing and validation will be extensive, to ensure healthcare information is being handled efficiently, accurately, and with the safeguards required for compliance with all standards of security and privacy.
- ✓ Training, support, and ongoing managed services will be provided as desired to ensure current success continues.

THE HEALTHLX CORE PLATFORM

The HealthLX Core Platform provides the basis for exchanging data between your systems and those of your healthcare delivery partners, including high-speed data transfer, data and format transformations, essentially unlimited scalability, easy customization of the data flows, high availability, and deep levels of granularity in data provenance and event tracking.

THE HEALTHLX CORE ENGINE

A powerful data integration platform designed to automate the flow of data between systems in real time. The engine empowers organizations to easily and securely move, transform, and process data across disparate sources, making it an essential tool for any data-driven business.

THE HEALTHLX FLOW EDITOR

A web-based flow builder with intuitive, clear interfaces to build simple or complex workflows by placing, connecting, and configuring available components. With its intuitive user interface and drag-and-drop functionality, no coding is required unless significant customizations are needed.

THE HEALTHLX MONITOR

A single-screen visualization of all flows running in an instance of the Platform, showing metrics and offering drill-down visibility of load, performance, statistics, and error information.

THE HEALTHLX EVENTS & NOTIFICATIONS MODULE

The Core platform includes deep, precise monitoring of all events and all data flowing through the system. The Events and Notifications module enables creating triggers and actions based on any of that information - business events, data content, attributes, system events, even expected events that did not happen - at any level desired. Dependency management is also provided, to prevent cascading messages from a root cause event.

PURPOSE-BUILT COMPONENTS™

Configurable pre-built components, requiring no coding, provide much of the functionality for your workflows.

Many of these components are provided as HLX products. Other users of the HealthLX Platform will contribute "community-built" components for customized needs. With appropriate licensing, your developers can also create new components to handle specific needs that arise.

All such components become available to be placed on workflow palettes as needed, and dropped into place in any flows that need them.

Interface standards such as HL7 always bring mapping issues to be resolved, differences between your existing data and the way the standard represents that information. Simple transformations are usually sufficient to resolve these, such as converting a number to an alphanumeric string of characters. When more complex transformations are needed, Purpose-Built Components can address those.

For a more sophisticated and customized PBC development, your development team can work to build custom PBCs, as new components for your dataflow composition.

SUMMARY

HL7 and FHIR®

The years of work by the HL7 foundation to establish clearly defined standards for sharing information, powered by the speed and reliability of today's communications tools, offer excellent opportunities to deliver better care, faster, at a lower cost, than ever before.
New compliance requirements will push everyone to implement these standards, and meeting the emerging deadlines will require significant work.

COMPOSABLE AUTOMATION™

As Gartner says, a composable architecture, as offered by the HealthLX Core Platform, is a "framework that enables organizations to rapidly build new, or extend existing, business capabilities, products and services. The framework is based on the ability to assemble and reassemble various digital assets and business elements for real-time adaptability and resilience in the face of uncertainty.

In your constant pursuit of improved patient care and reduced costs, getting maximum value from all available information is vital. The Composable Automation™ platform and tools offered by HealthLX accelerate your ability to achieve those goals.

The FHIR First™ orientation of the HealthLX team encourages rapid implementation of enterprise-level clinical data exchange between payers, providers, health information exchanges, EMR systems, and other participants. The excellent foundation of HL7, FHIR®, and the HLX Core platform, and the rapid assembly of new Purpose-Built Components for any additional needed functionality, accomplish that goal.

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